

T2DM Unplanned Admission Risk Prediction Model

August 2026 Clinical Review

Cohort:	300,290 patients with T2DM
Baseline risk:	7.47% will have an unplanned admission in the next 12 months
Current benchmark:	PR-AUC 0.427 ✓ (was 0.396 in July)
Gain this cycle:	+0.031 (+7.9%) — largest single-experiment gain in the project
Key finding:	Missed care (no nutrition counseling) is now the #1 risk predictor
August experiments:	2 completed · 1 PASS + new benchmark · 1 PASS + feature retained

What We'll Cover

- 1** **Headline result** Care Gap Indicators raised the benchmark to PR-AUC 0.427 — the project's biggest gain.
- 2** **Model in practice** What 5.9× lift means for care management, with updated operating points.
- 3** **A data correction done right** CGM coding was fixed and insulin pump use emerged as a distinct clinical signal.
- 4** **Utilization patterns** Care fragmentation and no-show behavior are measurable and ready for September.
- 5** **What comes next** The path to model lock and deployment readiness.

The Model Is a Prioritization Tool — Updated Operating Points

Operating strategy	Patients / cycle	PPV	Sensitivity	Lift
High-confidence (top 1%)	~900	~76%	~10%	10.1×
Balanced start (top 5%) ← Recommended start	~4,500	~44%	~30%	5.9×
Broad outreach (top 10%)	~9,000	~36%	~48%	4.8×
High-volume (top 20%)	~18,000	~26%	~69%	3.5×
Random selection	Any	~7.5%	Proportional	1.0×

How care teams use this

At top 5%, a care manager finds about 44 true high-risk patients per 100 reviewed—5.9× more productive than random review.

What changed vs. July

PPV rose from ~39% to ~44% at top 5%: about 5 more true high-risk patients per 100 reviewed, or ~225 fewer wasted contacts per 4,500-patient cycle.

START HERE · Use the top-5% threshold, then recalibrate after one live review cycle based on actual team capacity.

Patients Missing Expected Diabetes Care Are the Strongest Signal of Risk

+0.031

PR-AUC improvement (+7.9%)

0.427

New benchmark (was 0.396)

12 CARE-GAP INDICATORS

No A1C, eGFR, lipid panel, retinal exam, foot exam, nutrition counseling, diabetes education, plus aggregate gap count. The 22-month window exceeds annual screening cadence.

FOR RISK PREDICTION

The model better identifies patients falling through the cracks of structured diabetes management.

Metric	July	August
PR-AUC	0.396	0.427 ✓
PPV at top 5%	~39%	~44%
Sensitivity	~27%	~30%

#1 predictor:

no nutrition counseling in 22 months

- Fewer than 8% received nutrition counseling.
- Fewer than 9% received documented diabetes education.
- These gaps are common—and highly predictive.

FOR INTERVENTION DESIGN

The strongest features—nutrition counseling and diabetes education—are services that can be offered directly.

Missing care is directly actionable: offer flagged patients the services they are missing. It is both the strongest predictor and the clearest intervention target.

An Important Nuance: Sicker Patients Get MORE Monitoring

The raw data shows a counterintuitive pattern: patients who DO have gaps show lower admission rates — often dramatically so.

Care gap	Admission Rate (gap present)	Admission Rate (gap absent)
No nutrition counseling	5.5%	32.6%
No diabetes education	6.5%	17.5%
No A1C test	3.0%	9.7%
No creatinine test	1.3%	9.0%

CONFOUNDING BY INDICATION

Sicker patients receive more monitoring precisely because they are higher-risk. Among nutrition counseling recipients, admitted and non-admitted patients have virtually identical gap burdens (3.8 vs 3.9 out of 11) — what separates them is prior severity: 2× prior admissions, 2× ED visits, +47% charges.

WHAT THE +0.031 GAIN PROVES

If care-gap features were purely redundant with existing severity measures, they would add zero. The largest gain in the project's history demonstrates they carry genuinely new information — but that information operates in multivariate interaction, not as a simple rule.

The relationship is real but complex. What matters: the model is measurably better, and the interventions — offering missed services — are actionable regardless of mechanism.

A Data Correction Revealed a New Clinical Signal — Insulin Pump Use

In July, CGM use was reported as a meaningful predictor. A data-labeling correction led to a clinically useful discovery.

WHAT HAPPENED

Two insulin pump device codes had been miscounted as CGM sensors. Correcting the device list initially dropped performance—meaning the miscoding had been quietly helping.

WHAT WE DISCOVERED

Insulin pump use is a meaningful, distinct clinical signal. About 10 per 100 pump users were admitted versus about 7 per 100 non-pump users—a one-third higher rate consistent with greater clinical complexity.

RESOLUTION

Corrected the CGM list to sensors only, then built insulin pump use as its own feature. The model recovered and slightly exceeded the July benchmark; Clarity SME validation is in progress.

~10 vs. ~7

admissions per 100
pump vs. non-pump users

The July CGM finding stands. Transparent correction produced a more accurate CGM signal and revealed insulin pump use as a second, distinct signal.

How Patients Use Care Is Now Measurable — Ready for September

#2

MODEL IMPORTANCE RANK

n_distinct_specialties

14 care-pattern features

Built and validated across 300,290 rows with no nulls, then staged for September.

- Encounter patterns
- Scheduling behavior
- ED revisit patterns

Beyond visit counts, all five engagement-pattern areas are feasible with existing data.

Feature area	What it captures	Coverage
Visit spacing & regularity	Consistent or erratic care?	90%
No-show & cancellation behavior	Patient disengagement?	72%
ED bounce-back (72h / 30d)	Repeat acute care use	26% with ED
Department & specialty fragmentation	How spread across the system?	86%

FEATURE SET Visit spacing, variability, regularity, distinct specialties/providers, fragmentation ratio, no-show and cancellation counts/rates, left-without-being-seen, appointment lead time, and 72-hour / 30-day ED revisits.

28% of T2DM patients had ≥1 no-show—consistent with identifying care trajectories that may be becoming unstable.

The Path Forward: From Model Improvement to Deployment Readiness

With the benchmark at 0.427 and all experiments complete, the model development phase is closing. The path ahead is about clinical verification and production deployment.

Step	What it tests / requires	Status
Patient chart review	Clinicians review sample of flagged patients — confirms clinical alignment, informs threshold	Next step
Prediction delivery decision	Dashboard, Epic integration, or staged? Clinical workflow should drive the choice	Decision needed
Scoring infrastructure	H2O model packaged as scheduled scoring pipeline against live patient cohort	After decisions
Ongoing maintenance	Retraining cadence, drift monitoring, cost of upkeep (DS team estimate forthcoming)	DS scoping
Model lock	Feature set frozen after insulin pump code validation (Clarity team)	After model lock
Prospective Validation	Forward-looking cohort test on locked model before go-live	After model lock

AUGUST EXPERIMENT SCORECARD

Experiment	Result	Impact
A: Monitoring & Care Gaps	PASS — new benchmark	+0.03 PR-AUC · 12 features · #1 predictor
B: Care Utilization Patterns	PASS — 1 feature retained	+0.001 PR-AUC · 14 features for September

The model is ready for clinical verification. Next: patient chart review, then this group decides where predictions should live — that answer shapes the deployment timeline and effort.